

## Course Write-Up (for Courses other than External Degree Programmes)

Version 4.0

| Key items                                                                                                                                                                                                                                                                   | PEIs' Inputs                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Course Title:<br>1. If it is 100% e-learning course, to add "(e-learning)" at the end of the course title.<br>2. If the course is conducted in Mandarin, to add "(Mandarin)" at the end of the course title.<br>3. Include prefix "(MC)" for stackable certificate courses. | <b>Advanced Certificate in Agentic AI Coding and Architecting</b>                                                                           |
| Course Developer                                                                                                                                                                                                                                                            | <input checked="" type="checkbox"/> Self-developed <input type="checkbox"/> Externally-developed <input type="checkbox"/> Jointly-developed |
| Qualification to be Awarded upon Course Completion<br><i>Note: The name of qualification to be awarded should be the same as the course title.</i>                                                                                                                          | <b>Advanced Certificate in Agentic AI Coding and Architecting</b>                                                                           |
| Qualification to be Awarded by:                                                                                                                                                                                                                                             | <b>Tertiary Infotech Academy Pte Ltd</b>                                                                                                    |

Brief Description of Course (including learning objectives for each module)

This Advanced Certificate develops architect-level competency in designing, building, and operating reliable AI agent systems using Claude Code. Each module is consists of 10 sessions (9 teaching + 1 assessment), conducted live online for 3 days a week, 7:00 PM – 10:00 PM (3 hours per session).

*Please provide the information on module and its learning objective in the following format:*

| S/N | Module Title                                                | Learning objective                                                                                                                                                                                                         |
|-----|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <b>Module 1: Agentic Architecture &amp; Orchestration</b>   | This module develops the ability to design agentic systems — mastering the agentic loop and Messages API, multi-agent orchestration, hooks/programmatic enforcement, and session and workflow management.                  |
| 2   | <b>Module 2: Tool Design &amp; MCP Integration</b>          | This module builds skills in designing effective tools — strong tool descriptions, structured error responses, tool distribution and selection, MCP server configuration, and use of built-in tools.                       |
| 3   | <b>Module 3: Claude Code Configuration &amp; Workflows</b>  | This module develops mastery of Claude Code — CLAUDE.md hierarchy and configuration, custom commands and skills, Plan Mode and iterative refinement, and CI/CD integration and batch processing.                           |
| 4   | <b>Module 4: Prompt Engineering &amp; Structured Output</b> | This module develops advanced prompting — explicit criteria and instruction design, few-shot prompting, tool use for structured output, and validation-retry loops with multi-pass review.                                 |
| 5   | <b>Module 5: Context Management &amp; Reliability</b>       | This module builds production reliability skills — context optimization and positioning, escalation and error propagation, handling context degradation in extended sessions, and human review and information provenance. |

| Target Audience                                                                                                                                                                                                            | <b>Primary Target Audience</b> <ul style="list-style-type: none"> <li>• Software engineers and developers building AI-powered applications who want to architect agentic systems.</li> <li>• Solutions architects and technical leads designing AI/LLM integrations for their organizations.</li> <li>• DevOps / platform engineers integrating AI agents into CI/CD and automation pipelines.</li> <li>• Data scientists and ML practitioners moving from model usage into production agent design.</li> <li>• Technical product managers who need a working command of agent architecture to guide teams.</li> <li>• IT professionals and consultants seeking a specialized, career-ready AI architecture credential.</li> </ul>                                                                                                                                                                      |                 |                                                            |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------|-----------------|---------|---|------------------------|----|-----------------------|---|-----------|----|--|---|-----------|----|--|---|------------------------|-----|------------------------------------------------------------|---|-------------------------|----|----------------------------------|---|------------|----|-----------------------------------------|----------------------------------|--|--|------------|
| (For certificate and above qualifications only) – Potential Employment Opportunities upon Course Completion                                                                                                                | <p>Graduates of the Advanced Certificate in Agentic AI Coding and Architecting can pursue a wide range of roles in both the public and private sectors, including:</p> <ul style="list-style-type: none"> <li>• AI Engineer / AI Agent Developer</li> <li>• AI Solutions Architect</li> <li>• LLM Application Engineer</li> <li>• Conversational / Agentic AI Engineer</li> <li>• Machine Learning Engineer (LLMOps)</li> <li>• AI Integration Specialist</li> <li>• Automation Architect</li> <li>• Technical AI Consultant</li> <li>• Developer Experience / Platform Engineer (AI tooling)</li> <li>• AI Product Engineer</li> </ul> <p>These roles span industries such as finance, government, healthcare, technology, and telecommunications, with opportunities for advancement into senior and lead AI architecture positions.</p>                                                              |                 |                                                            |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| Does the proposed course curriculum follow a foreign or international curriculum through full-time programme at the Primary, Secondary, and/or High School levels (i.e., grades 1-12) including self-developed curriculum? | <input type="checkbox"/> Yes <input checked="" type="checkbox"/> No<br><br><i>If yes, please state the location of PEI's premise for the course to be conducted in:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |                                                            |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| <b>Course Details</b> <ul style="list-style-type: none"> <li>• Mode of Delivery</li> <li>• Duration</li> <li>• Class Frequency</li> <li>• Total Course Hours</li> </ul>                                                    | <input type="checkbox"/> Face-to-face <input type="checkbox"/> Blended <input checked="" type="checkbox"/> E-learning (100% live synchronous virtual)<br>Duration: approx. 4 months (part-time)<br>Class Frequency: 3 days per week (Mon, Wed, Fri) x 3 hours per day<br>Total Course Hours: 150 hours (part-time)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |                                                            |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| Course Hours Computation                                                                                                                                                                                                   | <p><i>Please provide the computation for course hours below (N/A if not applicable):</i></p> <table border="1"> <thead> <tr> <th>S/N</th><th>Course component</th><th>Number of Hours</th><th>Remarks</th></tr> </thead> <tbody> <tr> <td>1</td><td>Classroom Facilitation</td><td>NA</td><td>100% virtual delivery</td></tr> <tr> <td>2</td><td>Practicum</td><td>NA</td><td></td></tr> <tr> <td>3</td><td>Practical</td><td>NA</td><td></td></tr> <tr> <td>4</td><td>Synchronous e-learning</td><td>135</td><td>Live online teaching + hands-on labs (45 sessions x 3 hrs)</td></tr> <tr> <td>5</td><td>Asynchronous e-learning</td><td>NA</td><td>Optional self-study, not counted</td></tr> <tr> <td>6</td><td>Assessment</td><td>15</td><td>3 hrs assessment per module x 5 modules</td></tr> <tr> <td colspan="3"><b>Total Duration (in hours)</b></td><td><b>150</b></td></tr> </tbody> </table> | S/N             | Course component                                           | Number of Hours | Remarks | 1 | Classroom Facilitation | NA | 100% virtual delivery | 2 | Practicum | NA |  | 3 | Practical | NA |  | 4 | Synchronous e-learning | 135 | Live online teaching + hands-on labs (45 sessions x 3 hrs) | 5 | Asynchronous e-learning | NA | Optional self-study, not counted | 6 | Assessment | 15 | 3 hrs assessment per module x 5 modules | <b>Total Duration (in hours)</b> |  |  | <b>150</b> |
| S/N                                                                                                                                                                                                                        | Course component                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Number of Hours | Remarks                                                    |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| 1                                                                                                                                                                                                                          | Classroom Facilitation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NA              | 100% virtual delivery                                      |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| 2                                                                                                                                                                                                                          | Practicum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NA              |                                                            |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| 3                                                                                                                                                                                                                          | Practical                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NA              |                                                            |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| 4                                                                                                                                                                                                                          | Synchronous e-learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 135             | Live online teaching + hands-on labs (45 sessions x 3 hrs) |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| 5                                                                                                                                                                                                                          | Asynchronous e-learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | NA              | Optional self-study, not counted                           |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| 6                                                                                                                                                                                                                          | Assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 15              | 3 hrs assessment per module x 5 modules                    |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| <b>Total Duration (in hours)</b>                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 | <b>150</b>                                                 |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| • Minimum Entry Requirements                                                                                                                                                                                               | <p>Age: 21 years old</p> <p>Language Proficiency: At least C6 for GCE "O" Level English (or equivalent)</p> <p>Academic: At least a Diploma in IT / Computer Science / Engineering, OR GCE "A" Level, OR relevant work experience. Basic programming and command-line literacy required.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                            |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |
| (For certificate qualifications only) – to be                                                                                                                                                                              | Will the PEI issue modular certificate(s), that the students can stack towards the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |                                                            |                 |         |   |                        |    |                       |   |           |    |  |   |           |    |  |   |                        |     |                                                            |   |                         |    |                                  |   |            |    |                                         |                                  |  |  |            |

| read in conjunction with SSG Advisory 3/2022                                     | <p>award of the Certificate Qualification?</p> <p><input checked="" type="checkbox"/> YES   <input type="checkbox"/> NO</p> <p><i>Note: Only "Terminal" Certificate Courses can be registered as Modular Courses that can stack to a Certificate Qualification.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                             |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------|----|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                  | <p>All certificate qualification(s) that the modular course can stack to, and the module combinations that lead to that qualification:</p> <table border="1"> <thead> <tr> <th>S/N</th><th>Title of certificate qualification</th><th>Modules</th></tr> </thead> <tbody> <tr> <td>1.</td><td><b>Advanced Certificate in Agentic AI Coding and Architecting</b></td><td>           1. Agentic Architecture &amp; Orchestration<br/>           2. Tool Design &amp; MCP Integration<br/>           3. Claude Code Configuration &amp; Workflows<br/>           4. Prompt Engineering &amp; Structured Output<br/>           5. Context Management &amp; Reliability         </td></tr> </tbody> </table> | S/N                                                                                                                                                                                                         | Title of certificate qualification | Modules | 1. | <b>Advanced Certificate in Agentic AI Coding and Architecting</b> | 1. Agentic Architecture & Orchestration<br>2. Tool Design & MCP Integration<br>3. Claude Code Configuration & Workflows<br>4. Prompt Engineering & Structured Output<br>5. Context Management & Reliability |
| S/N                                                                              | Title of certificate qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Modules                                                                                                                                                                                                     |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
| 1.                                                                               | <b>Advanced Certificate in Agentic AI Coding and Architecting</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1. Agentic Architecture & Orchestration<br>2. Tool Design & MCP Integration<br>3. Claude Code Configuration & Workflows<br>4. Prompt Engineering & Structured Output<br>5. Context Management & Reliability |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
|                                                                                  | <ul style="list-style-type: none"> <li>The maximum amount of time allowed to complete the modular courses to be awarded the certificate qualification(s):<br/>approx. 4 months (part-time)</li> <li>The terms and conditions to be met so as to be awarded the Certificate:<br/>To be awarded the certificate, participants must:               <ul style="list-style-type: none"> <li>Achieve a pass in all required assessments (all 5 module assessments and the capstone), and</li> <li>Maintain a minimum attendance of 75% throughout the course.</li> </ul> </li> </ul>                                                                                                                         |                                                                                                                                                                                                             |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
| Articulation Pathway                                                             | <p>Is the proposed course a pathway programme?</p> <p><input type="checkbox"/> YES   <input checked="" type="checkbox"/> NO</p> <p><i>If yes, please furnish the supporting documents.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                             |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
| Information on External Course Developer (for externally developed courses only) | Not applicable — the course is self-developed by Tertiary Infotech Academy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                             |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
| Other Information, if applicable                                                 | N.A.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                             |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
| Course Accreditation                                                             | <p>Is the proposed course accredited by any external organisation?</p> <p><input type="checkbox"/> YES   <input checked="" type="checkbox"/> NO</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                             |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
| Association, Collaboration or Affiliation                                        | <p>Is there any other form of association, collaboration or affiliation with any other organisation or persons, either local or foreign, in respect of this course?</p> <p><input type="checkbox"/> YES   <input checked="" type="checkbox"/> NO</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                             |                                    |         |    |                                                                   |                                                                                                                                                                                                             |
| Courses with Industrial Attachment                                               | <p>Does the proposed course have industrial attachment?</p> <p><input type="checkbox"/> YES   <input checked="" type="checkbox"/> NO</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                             |                                    |         |    |                                                                   |                                                                                                                                                                                                             |

## Declaration

I, **Dr. ANG CHEW HOE**, (NRIC/Passport no. S1808997A) manager of **TERTIARY INFOTECH ACADEMY PTE. LTD.** declare that:

- (a) the course has been approved by the Academic Board;
- (b) the Academic Board has approved the deployment of teachers for the course in accordance with Regulation 26 of Private Education Regulations;
- (c) the assessment methods and procedures for the course has been approved by the Examination Board; and
- (d) all information provided in this application is true and accurate to my knowledge.

*Alfred Ang*

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(Signature of manager & date)

Date: 12 Jun 2026